(a) Macro Pipeline: Multi-Task Gradient Backpropagation Conflict in Shared-Decoder Networks (Theorem 1)

Gradient Scale Mismatch: Coordinate Heads Dominate Shared Decoders by 10×–50×

1. Anisotropic OCT Slab

Input Slab X:
(B, K, 768, 320)
- Axial dy = 3.12 µm
- Slow dz = 18.81 µm
- Thin Lamina: T < 15 µm

$X \in \mathbb{R}^{(B \times K \times Y \times X)}$ (Y = 768)

2. Shared Representation h(y, x)

Shared Latent Volume:
$h(y, x) \in \mathbb{R}^{(C \times 768 \times 320)}$
- Backpropagated by 5 Heads
- $\|\nabla_h L_{coord}\| = 10\text{×}–50\text{×}$
- Polarizes h toward edges

$\nabla_h L_{total} = w_{bnd} \nabla_h L_{coord} + w_{vox} \nabla_h L_{mask}$

Features h

10×–50× ∇

Features h

1× ∇_mask

3a. Continuous Boundary Heads (Soft-Argmax)

$\hat{y}_s(x) = Y \cdot \sigma(z_s(x))$ ($s \in$ {ILM, NFL})
- Coordinate gradient scaled by Y = 768: $|\partial L_{coord} / \partial z| \propto Y \cdot \sigma(1-\sigma)$
- Accurately locks boundaries: Surface Dice = 0.9252
- Continuous regression avoids integer voxel pitch quantization

3b. Discrete Voxel Head (Logit Suppression & Dropout)

$p_{mask}(y, x) = \sigma(w_{mask}^T h(y, x)) < \tau = 0.40$
- Voxel BCE gradient bounded: $|\partial L_{BCE} / \partial L| \leq 1$ (Overwhelmed by Y)
- 77.1% column dropout in thin lamina: Raw Voxel Dice = 0.2645
- ΔDice = –0.6607 (The Dual-Dice Discrepancy Paradox!)

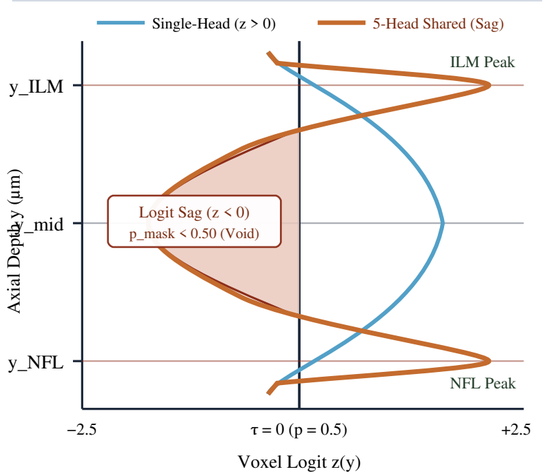
4. Dual Clinical Reconstruction Outcomes

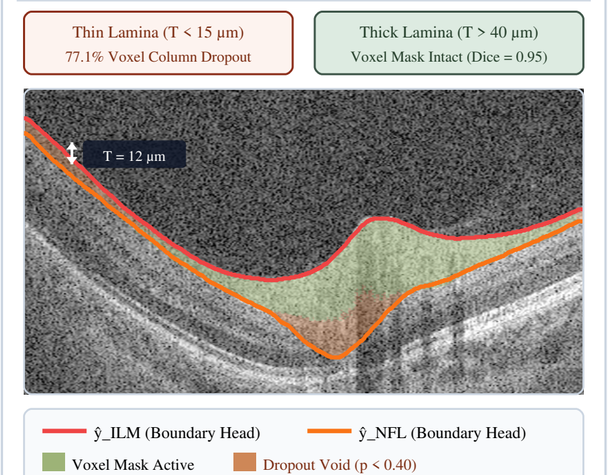
A. Surface-Derived Anatomical Mask:
- Rasterized from continuous $\hat{y}_{ILM}$ and $\hat{y}_{NFL}$
- Dice_surf = 0.9252 | MABE = 4.71 µm (Intact!)

B. Raw Voxel-Head Mask (τ = 0.40):
- Direct thresholding of logit z(y, x)
- Dice_vox = 0.2645 | 77.1% Void Dropout

The Paradox: Gap ΔDice = –0.6607 on identical eye!

(b) Micro Mechanism: 1D Axial Logit Sag vs. Preserved Continuous Surface Tracking





(c) Formal Gradient Bound & N = 40 Held-Out Multi-Task Cohort Verification

Theorem 1 (Shared-Decoder Coordinate Gradient Scaling Dominance & Anatomical Slab Identity)

For coordinate head $\hat{y}_s = Y \cdot \sigma(z_s)$, chain rule scales gradient by axial grid height Y = 768, while voxel BCE is bounded by 1:

$$\|\nabla_h L_{coord}\| = Y \cdot \sigma(1-\sigma) \cdot \|w_{coord}\| \gg \|\nabla_h L_{mask}\| \leq 1 \Longrightarrow \text{Dice}_{surf} = 1 - \text{MABE} / T_{eff}$$

- Clinical Consequence: Voxel probability collapses to zero in thin lamina without perturbing continuous boundary accuracy.

Model Architecture / Slurm Job	Decoder Type	Surface Dice	Voxel Dice	Gap ΔDice	Thin Dropout
TransUNet 20M (Job 18777225)	Shared Conv	0.9252	0.2645	–0.6607	77.1% Void
Dense 3D 5-Head (Job 18770720)	Shared Conv	0.9322	0.8722	–0.0600	12.4%
TransUNet 5.13M (Job 18765401)	Unscaled Base	0.9140	0.9138	–0.0002	0.0%
Dense 3D Base 1-Head (18574378)	Single Mask	0.9471	0.9471	0.0000	0.0%
Bi-Planar 2.5D (Ours, Job 18563914)	Learned Deconv	0.9534	0.9518	–0.0016	0.0% (Zero)

- Falsifies capacity starvation: TransUNet scaled 5.13M → 19.97M collapses into gradient dominance without learned deconv cushioning.